

HUZHUYUAN LONG

huzhiyuan.long@outlook.com · c-none.github.io · WeChat: miseinenn-lh

EDUCATION

The University of Hong Kong, *MSc* in Computer Science Sep 2025–Nov 2026
Tongji University, *B.Eng.* in Software Engineering Sep 2020–Jul 2024

PROFESSIONAL EXPERIENCE

NetEase Games — Knives Out Business Unit, Graphics Development Intern May 2026–Aug 2026

- Enhanced the ray-tracing component by building a path tracer as the engine's lighting-energy reference and adding emissive-light sampling.
- Extended GPU Lightmass to support partitioned baking for World Composition scenes.

Tencent — LIGHTSPEED STUDIOS, Game Engine Intern May 2025–Aug 2025

- Reimplemented SHaRC from RTXGI 2 in UE5 to replace Lumen's surface cache as a world-space radiance cache.
- Conducted controlled experiments to compare the two cache systems, quantify trade-offs, and guide radiance cache optimization.

PUBLICATIONS

SRSSIS: Super-Resolution Screen Space Irradiance Sampling for Lightweight Collaborative Web3D Rendering Architecture.

Huzhiyuan Long, Yufan Yang, Chang Liu, Jinyuan Jia. *CAD/Graphics (CCF-C)*. Aug 2023

- Proposed a super-resolution-based collaborative rendering architecture.

PROJECTS

WebGPU Ray Tracing

- Built a hybrid rendering pipeline on WebGPU with V-buffer, ray tracing, denoising, and super-resolution modules.
- Implemented an SAH-based BVH, reproduced ReSTIR DI & GI for real-time global illumination, and integrated SVGF and ReLAX for denoising.

Screen Probe Precomputation

- For a collaborative rendering system based on UE5 Lumen, delivered an early alternative interface for outputting screen probes to test the rendering pipeline.
- Implemented on a Vulkan hardware ray-tracing pipeline.

Precomputed Visibility Volumes

- In instanced Vulkan scenes, precomputed each sample point's visible-object set and weights to form per-cell visibility sets.
- Used visibility volumes to aid scheduling and streaming during real-time client-side model loading.

HONORS AND AWARDS

China Collegiate Programming Contest — Shanghai Collegiate Programming Contest, Bronze Medal Sep 2022

Group Programming Ladder Tournament, Individual Third Prize May 2022

Tongji University Undergraduate School-level Scholarship, Third Prize ×2 2022, 2021

TECHNICAL SKILLS

- Programming Languages:** C/C++, JavaScript, WGSL, GLSL, Python
- Frameworks and Tools:** Vulkan, WebGPU, UE, Three.js, Blender, Unity
- Languages:** English, Japanese (JLPT N2)